

Social Welfare under Heterogeneous Time Preferences

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Abstract. We study the synthesis of policies for stochastic systems acting on behalf of multiple principals with heterogeneous time preferences. We model this setting using asymmetrically-discounted Markov decision processes (MDPs), where each principal possesses an individual reward function and discount factor. The objective is to synthesize policies maximizing a utilitarian notion of social welfare, defined as the aggregate discounted payoff across all principals. Heterogeneous discounting fundamentally changes the structure of optimal policies: in contrast to classical discounted MDPs, welfare-optimal policies are generally not positional, even when all principals receive identical rewards. Nevertheless, we establish that optimal policies admit a simple structure: pure finite-memory counting strategies suffice and can be synthesized in polynomial time under mild assumptions on the spacing of discount factors. We further show that restricting synthesis to stationary strategies leads to computational intractability: deciding whether there exists a stationary strategy achieving a given welfare threshold is NP-hard, even for two principals with identical rewards. Our results highlight how heterogeneous temporal preferences introduce new synthesis challenges at the intersection of formal methods, multi-agent planning, and algorithmic game theory.

Keywords: Formal verification, validation and synthesis, Planning and Scheduling, Theoretical foundations of planning

1 Motivation

Many modern decision-making systems operate on behalf of multiple principals with differing preferences and priorities. Examples include AI-assisted resource allocation, climate-policy planning, autonomous economic systems, and collective decision-making infrastructures. In such settings, principals may evaluate future outcomes differently due to distinct temporal preferences.

Discounted Markov decision processes (MDPs) provide a canonical framework for sequential decision-making under uncertainty. Classical discounted MDPs assume a single discount factor shared across all rewards. However, this assumption fails to capture heterogeneous notions of patience among multiple stakeholders.

We study the synthesis problem for *asymmetrically-discounted MDPs*, where each principal is associated with an individual reward function, and an individual discount factor.

The system acts according to a single shared policy, and the objective is to maximize a utilitarian notion of social welfare obtained by aggregating discounted utilities across all principals. Our work is motivated by observations from behavioural economics and cognitive science, such as the Marshmallow Experiment, which demonstrates that individuals differ substantially in their willingness to delay gratification. These heterogeneous time preferences naturally induce conflicting views on long-term planning and raise a synthesis question:

How should a system act when different principals value future rewards differently?

A central phenomenon emerging from heterogeneous discounting is that optimal social welfare may require memory. Consider two principals jointly controlling a system. One principal may prefer immediate rewards while another values long-term gains more heavily. In such cases, welfare-optimal behaviour may initially favour the impatient principal before eventually switching to a policy preferred by the patient principal. Consequently, optimal strategies are generally not positional.

Contributions. Our main contributions are:

- We introduce asymmetrically-discounted MDPs as a formal model for social-welfare synthesis under heterogeneous time preferences.
- We show that welfare-optimal policies are, in general, not positional, even when all principals receive identical rewards.
- We prove that optimal social welfare can nevertheless be achieved using pure finite-memory counting strategies.
- We present a polynomial-time synthesis algorithm under mild assumptions on the spacing of discount factors.
- We show that restricting synthesis to stationary strategies leads to NP-hardness, both for pure and randomized stationary strategies.

2 Model and Algorithm

An asymmetrically-discounted MDP is a tuple (S, A, T, P, R, Λ) , where:

- (S, A, T) is a finite MDP,
- $P = \{0, \dots, n-1\}$ is a set of principals,
- $R : S \times A \times P \rightarrow \mathbb{Q}$ assigns rewards to principals,
- $\Lambda = \{\lambda_p \mid p \in P\}$ assigns discount factors $\lambda_p \in (0, 1)$.

For a strategy σ , the payoff for principal i is $D_i^\sigma(s) = \mathbb{E} \left[\sum_{j=0}^{\infty} \lambda_i^j R(s_j, a_j, i) \right]$.

We define the social welfare of σ as $SW^\sigma(s) = \sum_{i=0}^{n-1} D_i^\sigma(s)$.

The synthesis objective is to compute a strategy maximizing social welfare.

3 Synthesis of Welfare-Optimal Policies

Our synthesis procedure is based on the observation that optimal long-term behaviour can be characterized by a positional asymptotic policy.

We first solve the MDP from the perspective of the most patient principal. Whenever multiple actions remain optimal, we lexicographically refine the policy using increasingly impatient principals as tie-breakers. This produces a long-term policy that is eventually welfare-optimal. However, immediate deviations from this asymptotic policy may improve social welfare. We therefore analyze temporary deviations and show that such deviations become unprofitable after a bounded number of steps. This yields a decomposition of optimal behaviour into a finite deviation phase which is followed by a fixed asymptotic positional policy. The finite deviation phase can be synthesized by unfolding the MDP into a finite directed acyclic graph and solving a finite-horizon optimization problem. Welfare-optimal policies can be synthesized in polynomial time.

4 Hardness of Stationary Synthesis

Although unrestricted welfare synthesis is tractable, the picture changes dramatically when restricting attention to stationary strategies.

We show that deciding whether there exists a stationary strategy achieving a given social welfare threshold is NP-hard. The reduction proceeds from 3-SAT and holds even under strong restrictions: only two principals; identical rewards for all principals; or zero-sum rewards.

This creates a strikingly different situation compared to the case of uniform discounting: restricting strategies to simpler stationary policies loses welfare and increase computational complexity.

5 Conclusion

We introduced a synthesis framework for stochastic systems operating under heterogeneous time preferences. Our results show that temporal asymmetry fundamentally changes the structure of optimal decision-making policies.

While classical discounted MDPs admit positional optimal strategies, heterogeneous discounting requires finite memory even in simple cooperative settings. Nevertheless, optimal welfare synthesis remains tractable and admits structurally simple counting strategies.

The research we will present will appear in IJCAI 2026. A full version of this work is available on ArXiv [1].

These results open several directions for future work, including:

- quantitative fairness notions beyond utilitarian welfare,
- Pareto-optimal synthesis,
- stochastic games with strategic principals,
- learning discount factors from observed behaviour,
- and synthesis under partially observable preferences.

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